DBT PORTFOLIO PROJECT

LAYER: CONFIGURATION

DBT Source Definition

YAML | sources.yml | Data Lineage | Column Documentation | Source Testing

Project Summary

Declares and documents the external healthcare claims source table that all downstream DBT models reference. Defines the database, schema, table name, and all 14 column names -- enabling automatic lineage tracking, source freshness checks, and the {{ source() }} Jinja macro used across every model in this project.

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1 PROJECT OVERVIEW

File:	sources.yml (inside models/ directory)
Layer:	Configuration -- Source Declaration
Source Table:	healthcare_db.healthcare_schema.claims
Row Volume:	~50,000 rows (healthcare claims time series)
Columns Documented:	14 columns across time series, financials, and quality flags
Audience:	DBT developers, Data Engineers, Analysts

2 BUSINESS CONTEXT

Without a sources.yml declaration, dbt models that reference the claims table do so blindly -- no lineage, no documentation, no freshness checks. By declaring the source here, every {{ source('healthcare','claims') }} call in downstream models is automatically tracked in the dbt DAG, enabling the data graph view and source-level testing.

- Enables dbt source freshness checks -- alerts when the claims table is not updated on schedule.
- All 14 columns documented for auto-generated dbt docs site.
- Centralizes the source reference -- if the table moves schemas, only this file changes.
- Downstream models reference {{ source('healthcare','claims') }} -- never hardcoded table names.

3 COLUMN REFERENCE

Column	Type	Description
entity_id	VARCHAR	Unique entity identifier (provider, payer, or patient)
payer	VARCHAR	Insurance payer name
state	VARCHAR	State of claim origin
specialty	VARCHAR	Medical specialty of the claim
period_date	DATE	Reporting period date (time series key)
metric_value	FLOAT	Primary claim metric value
claim_volume	INT	Number of claims in period
return_value	FLOAT	Financial return metric for risk models
denial_rate	FLOAT	Claim denial rate (0-1)
days_to_payment	INT	Days from claim submission to payment
is_fraud_flag	INT	Binary fraud indicator (0/1)
y_value	FLOAT	Regression target variable
x1_value	FLOAT	Regression predictor 1
x2_value	FLOAT	Regression predictor 2

4 FULL SOURCE DEFINITION FILE

```yaml
version: 2

sources:
  - name: healthcare
    database: healthcare_db
    schema: healthcare_schema
    tables:
      - name: claims
        description: "Healthcare claims time series - 50k rows"
        columns:
          - name: entity_id
          - name: payer
          - name: state
          - name: specialty
          - name: period_date
          - name: metric_value
          - name: claim_volume
          - name: return_value
          - name: denial_rate
          - name: days_to_payment
          - name: is_fraud_flag
          - name: y_value
          - name: x1_value
          - name: x2_value
```